# Improving Training Instance Quality in Aerial Image Object Detection With a Sampling-Balance-Based Multistage Network

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***Abstract*— Object detection, aiming to recognize and locate objects of interest in aerial images, has historically played a significant role in the remote sensing community. Following remarkable improvements in Earth observation technologies, high-resolution remote sensing (HRRS) images with a bird's eye view perspective have revealed many categories of objects with sufficient variations in appearance and on complex backgrounds that make HRRS object detection an active but challenging task. The selection of positive samples and negative training instances is an essential factor in influencing detectors' performance. Related studies have found that many low-quality negative samples in the detectors' training process have caused training instability and low detection accuracy. In this work, a novel sampling-balance-based multistage network (SB-MSN) is presented to adaptively mine high-quality positive and negative instances for training an accurate detector. It has a series of components to ensure the selection and generation of high-quality examples for training an accurate detector, including a multiscale information retention module, an intersection over union balance sampling strategy, a balance L1 loss, and a multistage network. The proposed detector has been evaluated on three representative HRRS data sets. The extensive experimental results show that our detector can solve the problem of low-quality samples and significantly improve the detection performance of the mAP by 1.4% with the NWPU VHR-10 data set, 3.5% with the high-resolution remote sensing detection (HRRSD) data set, and 4.2% with the detection in the optical remote (DIOR) data set.**[1]

***Index Terms*— High-quality example mining, high-resolution remote sensing (HRRS), multistage detector, object detection, remote sensing.**

## I. Introduction

OBJECT detection in aerial images aims to locate objects of interest [e.g., vehicles (VEs) and airplanes (ALs)] in the image and identify their categories. The "object" generally refers to human-made objects [e.g., VEs, buildings, and ships (SHs)] that are independent of a complex background environment, as well as landscape, such as land-use/land-cover (LULC) parcels. With the development of observation technology in recent years, the spatial resolution of remote sensing images has greatly improved. For instance, the images extracted from Google Earth (Google Inc.) [1] and the WorldView-3 (DigitalGlobe Inc.) [2] are with resolution up to about 0.3 m. Compared with low-resolution images, high-resolution remote sensing (HRRS) images show more shape, color, texture features of objects of interest, and sufficient background information. Therefore, HRRS images are gradually becoming one of the main data sources for fine Earth observation [3], [4], which facilitates a wide range of applications. HRRS images are also applied in object detection.

The mainstream detection frameworks based on convolutional neural networks (CNNs), such as the single-shot MultiBox detector (SSD) [7], regions with CNN (RCNN) [8], Fast RCNN [9], and you only look once (YOLO) [10], have been designed for processing natural images. Some comparisons between natural images and HRRS images are provided in Fig. 1, showing that HRRS images contain many differences compared from natural images, which can be summarized as follows:

1) Natural images generally contain several categories and a few instances of objects. In contrast, an HRRS image covers an extensive range of Earth's surface and contains many kinds of objects. Sufficient variations of sensor, scale, color, shape, and texture are in a single-object category. Parts of objects are quite small; for example, VE is less than 25 pixels in width in an HRRS image. Many small objects are usually crowded into a specific region of aerial images.

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[1]Code is available at https://github.com/weihancug/Sampling-Balance_Multi-stage_Network

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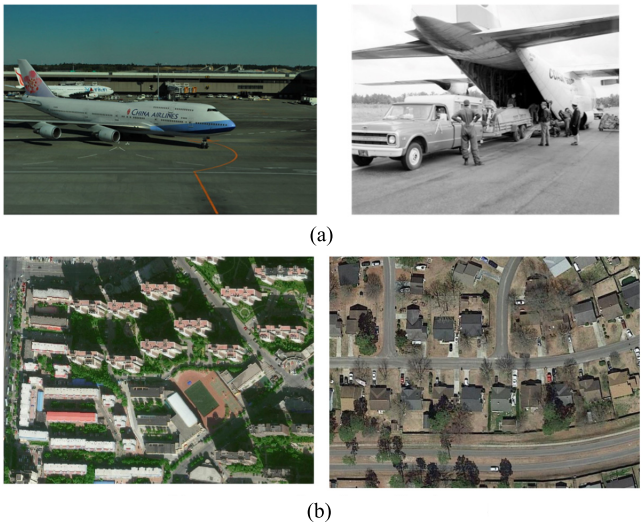


Fig. 1. Image examples of the COCO data set [5] and the HRRSD data set [6]. (a) Two examples of COCO data set. (b) Two examples of HRRSD data set.

2) HRRS images are typically taken from a bird's eye view, which implies that many types of objects of interest and the background form complex spatial patterns. The intricate patterns increase the difficulty of object detection in HRRS images.
3) Affected by weather, light, imaging conditions, and the shadows of surrounding objects, HRRS images are noisy. In this case, the color and texture of objects can easily appear to be changed.

Due to the bird's eye view perspective, highly complex backgrounds, and variant appearances of objects, particularly smaller ones, object detection in HRRS images is an active but challenging task in the field of remote sensing. Much effort has been put into developing outstanding models [11]–[19] and to create benchmarks [20]–[23]. The previous mainstream approaches can be categorized as template matching [24], prior knowledge, object-based image analysis (OBIA), and machine learning (ML) [25]. After 2012, deep learning (DL)-based detectors, as a branch of ML methods, have dominated and achieved remarkable progress in computer vision. Therefore, more and more scholars have been committed to apply DL-based detectors and continuously improve them to address the abovementioned challenges in remote sensing object detection. To detect multiscale geospatial objects, Qiu *et al.* [26] designed an adaptive aspect ratio multiscale network ($A^2$-RMNet), which utilizes several modules to capture discriminative features and automatically choose the appropriate aspect ratios for the anchors to match objects. To adapt to sufficient object deformations, such as scaling and rotation, Wu *et al.* [27], [28] developed the Fourier-based methods to extract the objects' rotation-invariant features. Dong *et al.* [29] applied transfer learning to develop a novel detector for addressing the problem of insufficient training samples in remote sensing object detection. Because the proper use of local and global context is beneficial for the improvement of object detection accuracy, Zhang *et al.* [30] presented a context-aware detection network (CAD-Net) to capture the correlations of local neighboring objects and the global information of the objects of interest.

Different from the abovementioned methods, the selection of positive samples and negative samples for reasonably training a detector is another essential factor in detector performance. Generally, prediction proposals that use intersection over union (IoU) and ground truth (GT) with a threshold greater than 0.5 are put forth as positive samples for training. However, positive samples are very few, and if the detector is trained only on positive samples, it may cause overfitting. If the IoU threshold of proposals is less than 0.5, they are set as negative samples. Negative samples are also important for training an accurate detector due to the limited quantity of positive samples. Nevertheless, some negative samples are of low-quality, with an IoU value less than 0.05, which may cause an unstable training process and a decline in detection accuracy. Some relevant studies [31] have revealed that most samples in training instances (accounting for more than 70%) are this type of low-quality samples, and they contribute the majority of gradients in the training process of detectors. Therefore, reducing the number of these types of samples is beneficial for boosting detector performance. Due to the large scale of HRRS images and the small scale of objects in those images, the problem of lack of high-quality samples is common and severe. Object features are weak and susceptible to interference from the background. In this case, proposals that are selected by a region proposal network (RPN) contain a large number of low-quality samples, and there are not enough high-quality positive and negative samples to train an ideal detector.

To efficiently and effectively mine high-quality training instances, we present a new sampling-balance-based multistage network (SB-MSN) to complete the task of object detection in aerial images. The SB-MSN has a series of components to improve the quality of training samples. Initially, given an input image, a multiscale information retention step is used to build multiscale pyramid images and then construct pyramid features for each of the pyramid images. It can preserve as much as possible of the multiscale information, and an RPN is used to identify a number of proposal candidates according to the pyramid features. Then, an IoU balanced sampling method is used to adaptively select high-quality negative samples for subsequent training. In the regression process, a balance L1 loss is utilized to replace with the smooth L1 loss for stabilizing the detector's training process. A multistage network consisting of a series of detection heads with increasing IoU thresholds is applied to locate and recognize objects, which can progressively generate more positive proposals for accurate predictions. Extensive experiments were conducted to evaluate the proposed method on three representative benchmark data sets. The results proved that the proposed detector improved detection accuracy (mAP) of 1.4% on the NWPU VHR-10 [21], 3.5% on the high-resolution remote sensing detection (HRRSD) [6], [32], and 4.2% on the DOIR [22]. The contributions of this study are summarized as follows.

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1) Focusing on the problem of insufficient high-quality training samples in remote sensing object detection, a novel end-to-end SB-MSN detector is designed to simultaneously select and generate high-quality training samples to boost the detection performance.
2) In the proposed detector, a multiscale information retention module is developed to preserve the scale information of an input image; both an IoU-balance sampling strategy and a multistage detection head can adaptively select and generate high-quality negative and positive training instances; and a balanced L1 loss is applied to stable the whole training process. The extensive experiments demonstrated the positive effect for each component and the superiority of the detector.

The remainder of this article is organized as follows. In Section II, we briefly review the related studies. The details of the proposed detector are described in detail in Section III. The experimental results and analysis are provided in Section IV. Finally, our conclusions are drawn in Section V.

## II. Related Work

### A. Mainstream Object Detection Methods

Remarkable progress has been made in feature representation and object recognition for object detection in the past several decades [22], [25], [33]–[36]. In the 1980s and 1990s, methods based on template matching [37]–[39] and prior knowledge [40], [41] were two categories of simple approaches to object detection. The template matching methods accomplished object detection by finding the best matches in an input image by using a series of predefined templates. Knowledge-based methods can transfer object detection into hypothesis testing problems by constructing a series of knowledge rules, wherein the establishment of knowledge and rules is the most important step. Later, with increasing accessibility of high-resolution images, OBIA methods were presented for mapping imagery into meaningful targets using image segmentation [42], [43]. Satisfactory performance can be achieved by selecting the desired scale, shape, and compactness criteria. These three methods are simple and easy to implement, and they achieve fast detection speeds when addressing a specific task. However, massive expert knowledge and human labor are needed to design them. They also have limited capabilities for processing complex, changeable, and large-scale remote sensing imagery. These characteristics hindered their widespread use in remote sensing.

Since the 2000s, with remarkable advances in ML methods, the capabilities of feature representation and classification have been dramatically improved. These newer techniques have achieved significant improvements in object detection. ML-based methods transfer the task of detection to classification by assigning a set of input regions the corresponding predicted labels. The most essential roles in the process are feature extraction with, for example, bag-of-words (BoW) feature [44], texture features [45], sparse representation (SR)-based features [46], feature fusion, dimension reduction (which is optional), classification using a support vector machine (SVM) [47], AdaBoost [48], k-nearest-neighbor (kNN) [49], or conditional random field (CRF) [50]. ML-based methods can automatically capture the variations in the appearance of objects, and their scalability and compatibility are improved. However, the representation ability is not sufficiently robust to deal with the object appearance variations in large-scale data sets.

After 2012, a branch of ML called DL attracted widespread attention and achieved significant progress in computer vision representation [51], [52]. DL models, especially CNNs [52]–[55], can extract high-level feature representations of an input image and boost classification accuracy. Girshick *et al.* [8] took the lead, applying CNNs to object detection by developing RCNNs. Since then, many milestone works have presented developments in object detection at an unprecedented speed. The DL-based detectors can be categorized into two classes according to the presence or absence of a proposal generation stage: two-stage frameworks and one-stage frameworks [56], [57].

1) Two-stage detection frameworks contain a preprocessing component for generating object proposals. These detectors have dominated in object detection since RCNN [8] is due to their remarkable detection performance on benchmark data sets. The milestone models are RCNN [8], spatial pyramid pooling network (SPPNet) [58], Fast RCNN [9], Faster RCNN [59], and region-based fully convolutional network (RFCN) [60].
2) A one-stage detector directly predicts the locations of the bounding box and the class probabilities in an entire image using a single CNN. It does not involve region proposal generation, feature resampling, or post-classification, encapsulating all computation in a single network. The representative models are YOLO [10], [61], [62], SSD [7], and RetinaNet [63]. Due to encapsulating all computation in a single network, one-stage models are computationally effective, but the detection performance is slightly worse than the two-stage models.

### B. Methods for Handing Insufficient High-Quality Training Instances

The methods to address insufficient high-quality instances during detectors' training process can be classified into three categories according to the levels that they work on: annotation data, feature representation, and training proposal (or instance). In the level of annotation data, increasing training samples can effectively increase high-quality proposals during the training process. However, labeling annotation samples is a labor-extensive and knowledge-consuming task. Therefore, data augmentation, as an alternative technique, is commonly used by rotating, resizing, and flipping original samples. Transfer learning [29], active learning [64], semisupervised or weak-supervised learnings [65]–[69], and generative adversarial net (GAN)-based sample generation [70], [71] are another four representative kinds of methods to augment the training samples. Transfer learning is an ML technique where a model trained on a task is reused on another task, which can speed up training and improve the performances of DL models. Dong *et al.* [29] presented a Sig-non-maximum suppression (NMS)-based faster R-CNN with transfer learning

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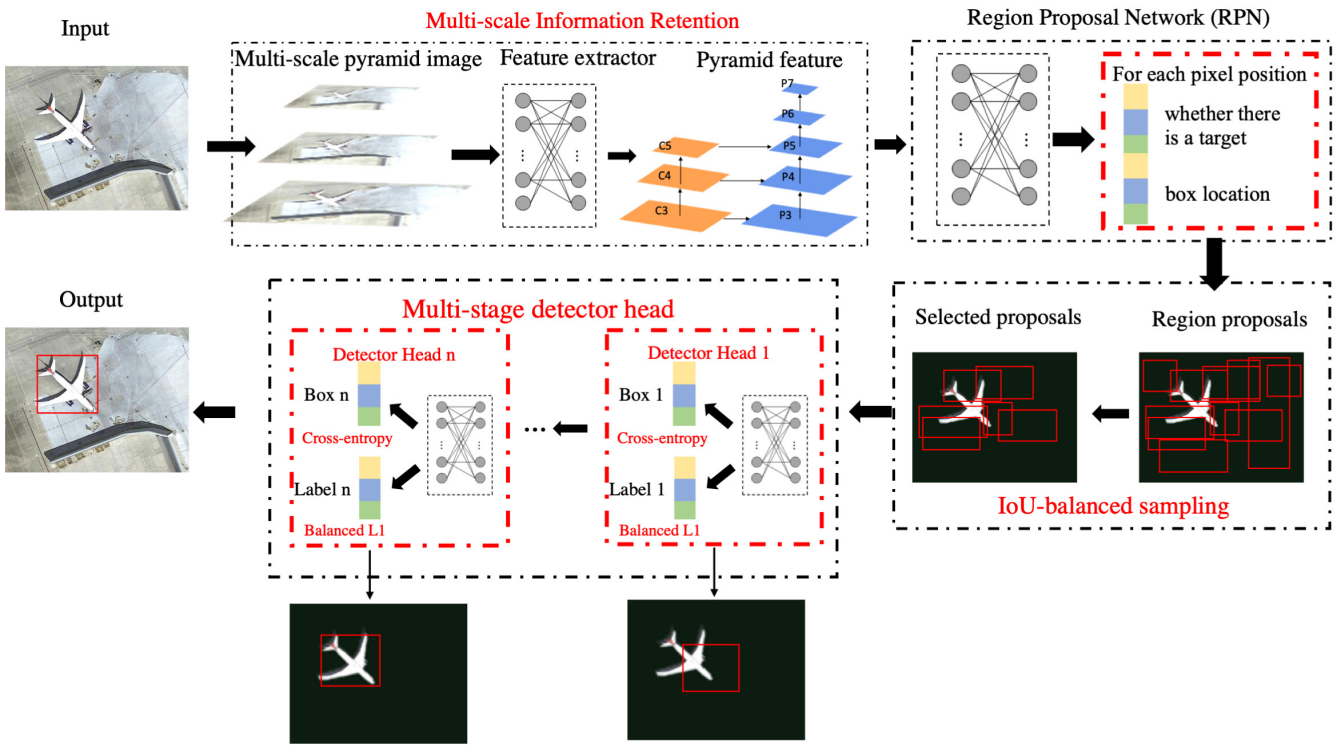



Fig. 2. Overall architecture of the proposed detector.

for improved detection performance. Active learning is a strategy in which a learning model can interactively query or select valuable samples for labels, which can reduce the requirements of the number of annotation samples. Chan-Hon-Tong and Audebert [72] and Kellenberger *et al.* [64] used active learning to collect highly confident samples for quick retrieval of true positives in the training data set. Cao *et al.* [73] developed a unified framework by integrating both DL and active learning to reduce labeling cost. Semisupervised and weak-supervised learnings can make a model, trained on a small number of correct or weak annotation samples, to iteratively mini-valuable samples from the unlabeled data. Zhang *et al.* [65] used an iterative weakly supervised learning framework to automatically select and augment the training data. Due to the remarkable ability of modeling distributions and high-quality image generation of GANs, they are used for data augmentation. Han *et al.* [71] developed a supervised Wasserstein GAN to generate annotation samples for scene classification. The methods of the feature representation level are achieved by strengthening feature discrimination to select valuable samples, which is often combined with semisupervised learning, active learning, and multimodal data [67], [74].

Positive sample mining is another essential technique for finding important samples in the level of training proposal. Research in the field of remote sensing has revealed that existing detection data sets contain an overwhelming number of easy instances and only a few hard examples. Therefore, many scholars have tried to mine more representative hard examples to balance the proportion of foreground-background class instances. Cheng *et al.* [75] presented a two-step iterative training method that alternates between updating the detection model given to the training set and adaptively selecting the hard negative examples for updating the detection model. Different from the abovementioned methods, we developed an end-to-end SB-MSN detector to generate high-quality examples that boost detection performance. By doing this, the detector is effective at detecting hard instances.

## III. Methodology

In this section, we apply a series of modules to construct our unified SB-MSN, which can progressively improve the quality of training samples for achieving satisfactory detection performance. Fig. 2 shows the whole processing flowchart of the proposed detector, which includes the following.

1) *Multiscale Information Retention Module:* Given an input image, it first constructs a series of pyramid images from high resolution to low resolution according to a predefined rule. For each one of the pyramid images, this module then applies a feature extractor to learn and generate multiscale features. Therefore, the objects can be detected at an appropriate scale, which is beneficial to recognize and locate them. The abovementioned two operations are convenient for retaining as much multiscale information as possible, especially with overly large or small objects.
2) *IoU-Balanced Sampling Strategy for High-Quality Example Mining:* The extracted pyramid features are fed to an RPN to generate a large number of proposal candidates. Due to numerous low-quality negative samples (their IoU values with the GT less than 0.05) in the proposal candidates, an IoU balanced sampling

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method is used to select high-quality negative samples for subsequent training adaptively.

3) *Balance L1 Loss for Smoothing Training Process:* In the regression process, some outlier samples can generate tremendous loss value in the localization branch of the detection head, which is critically harmful to the overall training degree. For this reason, we replaced the smooth L1 loss with a balance L1 loss, which can significantly reduce the influence of the unbounded regression targets during the training process. By doing this, the training process of the detector will be more stable.
4) *Multistage Detection Network for Positive Sample Generation:* Although high-quality negative samples have been increased through the abovementioned operations, there are still relatively few positive samples in the training process. Therefore, we constructed a multistage network encompassing a series of detectors with increasing IoU thresholds, which can progressively improve the accuracy of prediction proposals. In this case, positive samples significantly increase for accurately training subsequent detection heads.

The following four subsections detail each component of the proposed SB-MSN detector.

### A. Multiscale Information Retention Module

Significant differences in object scales, sufficient variations in object appearances, complex backgrounds, and overlarge images make HRRS object detection a more challenging task. Some categories of objects, such as football fields and harbors, occupy more than 300 pixels in width in an image. For these kinds of large-size targets, the challenge for the detector is to extract global information, which results in the inability to recognize and locate targets correctly. Meanwhile, some small-scale targets, such as VE and small SHs, are less than 3-m wide, and their color, shape, and texture information are weak and easily changed. An appropriate scale of an input image or feature can be beneficial for correctly identifying the category and location of a target.

In order to robustly detect multiclass objects, we applied two techniques for constructing a multiscale information retention module that can preserve as much multiscale information from the targets as possible via the following steps.

1) In this module, the resize ratios $R$ for constructing pyramid images are set as $R = \{1.5, 1, 0.8, 0.6, 0.3\}$. An input image is first enlarged or reduced according to the predefined ratios $R$ to obtain a series of images from high resolution to low resolution. High-resolution images are suitable for detecting small targets, and low-resolution images are available for large targets. These images together form the pyramid images for the input image, each of which would be fed to the subsequent parts of the model.
2) For the pyramid images, a deep CNN model, such as ResNet [54] or ResNeXt [55], is set as the feature extractor, which is pretrained on ImageNet [51] and initially applied in large-scale classification tasks. Due to the excellent feature representation ability, the deep CNN model is set as a backbone to extract high-level features in object detection. The CNN-based feature extractor further integrating feature pyramid network (FPN) [76] can learn and preserve multiscale features. Among them, low-level features preserve the location information of targets, and high-level features retain semantic information.

The two steps work together to ensure that targets of interest can be trained and detected with the most suitable scale information, which enhances robustness and stability.

### B. IoU-Balanced Sampling Strategy for High-Quality Example Mining

The multiscale pyramid features are input into the RPN to generate a large number of proposal candidates for the regression training. However, there is a critical problem in the proposal candidates obtained with random sampling in RPN. The relevant study [31] reported that more than 60% of the hard proposal candidates had an overlap ratio with GT greater than 0.05. This type of proposal is the key to improve the performance of a detector. However, the RPN in the manner of random sampling strategy would generate 70% candidates with the overlap ratios less than 0.05, which is not conducive to obtaining an ideal model. Therefore, an IoU-balanced sampling strategy is applied in our detector to mine more hard negative candidates as the training instances.

For the given $M$ proposal candidates in an region of interest (ROI) pooling layer, $N$ samples are selected for training the detection head. The selection probability of each sample in random sampling is

$$p_r = \frac{N}{M}. \tag{1}$$

To increase the selection probability of hard examples, the candidates with higher IoU values must be selected. Therefore, the proposal candidates are divided into $K$ groups according to IoU values. $N$ samples are equally distributed to every group. Then, $N$ samples are selected under the abovementioned strategy according to

$$p_k = \frac{N}{K} * \frac{1}{M_k} \tag{2}$$

where $M_k$ is the number of selected proposal candidates in each group. $K$ is set to 3 by default [31]. The sampling method makes the detector more likely to choose high-quality hard samples for the following regression training.

### C. Balance L1 Loss for Smoothing Training Process

The classification and localization tasks of a unified detector can be addressed simultaneously with a multitask loss [9], which can be formulated as

$$L_{p,u,b^u,v} = L_{\text{cls}}(p, u) + \lambda[u \geq 1]L_{\text{loc}}(b^u, v) \tag{3}$$

where $L_{\text{loc}}$ and $L_{\text{cls}}$ are two independent objective functions corresponding to the localization and recognition objects, respectively. For an object, the prediction label and GT are denoted as $p$ and $u$; $b^u$ is the coordinates of prediction location

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with class $u$; $v$ is the regression target of the location; and $\lambda$ is the weight coefficient for balancing the loss effects of classification and localization tasks.

In the aspect of $L_{\text{cls}}$, the classifier is the function $p = C(I)$ that assigns an input $I$ to one of $O + 1$ predefined classes, where the category "0" is background and the remaining classes of the objects of interest to detect. Given a training set $(I_i, u_i)$ where $I$ is the training samples and $u$ is the label, this is learned by minimizing the cross-entropy loss

$$R_{\text{cls}}[C] = \sum_i -\log C_u(I). \tag{4}$$

In the aspect of $L_{\text{loc}}$, a bounding box $(c_x, c_y, w, h)$ has four coordinates of an input $I$. Bounding box regression is to regress a prediction bounding box $b^u$ into a GT bounding box $v$, using a regressor $f(I, b^u)$. It is learned from a training set $(b^u, v)$, by minimizing the smooth $L1$ loss [9]

$$L_{\text{loc}}(b^u, v) = \sum_{j \in (c_x, c_y, w, h)} \text{smooth } L1\left(b_j^u - v_j\right) \tag{5}$$

where

$$\text{smooth } L1(x) = \begin{cases} 0.5x^2, & |x| < 1 \\ |x| - 0.5, & \text{otherwise.} \end{cases} \tag{6}$$

However, due to many unbounded regression targets in the proposal examples, some outlier objects would likely produce a huge location loss, which would have contributed 70% gradients to the overall gradients and cause the unstable training process [31]. It is unreasonable to directly balance the classification and localization loss for training the detector. Therefore, we applied a balance L1 loss in this study to stabilize the training process of the detector.

The main idea of the balance L1 loss is promoting the gradients of the normal and hard regression targets, which can achieve a more balanced training degree within classification and localization task, which is defined as

$$L_b(x) = \begin{cases} \frac{\alpha}{b}(b|x| + 1)\ln(b|x| + 1) - \alpha|x|, & \text{if } |x| < 1 \\ r|x| + C, & \text{otherwise} \end{cases} \tag{7}$$

where the three parameters $\gamma$, $\alpha$, and $b$ are constrained by

$$\alpha \ln(b + 1) = \gamma. \tag{8}$$

The parameters $\alpha$ is set to 0.5, and $\gamma$ is set to 1.5 by default, respectively. The balance L1 loss increases the gradients of normal samples under the control of $\alpha$. A small value of $\alpha$ improves the gradients of the normal samples, but the gradients of outliers are not influenced. A $\gamma$ is used to adjust the upper regression bounds of outliers, which can help the overall objective function to be more stable. Finally, $b$ is applied to ensure that the two formulations in (7) have the same value.

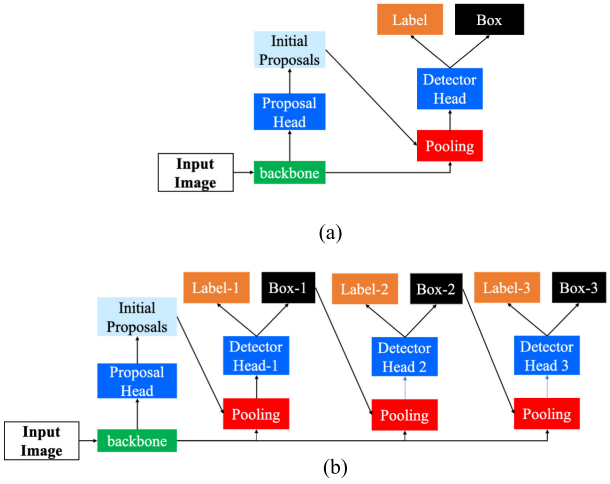



Fig. 3. Comparison for (a) two-stage detector [59] and (b) multistage detector [77], [78].

### *D. Multistage Detection Network for Positive Sample Generation*

Fig. 3 shows the two-stage [59] and multistage models [77], [78], in which the input image is processed by a feature extractor, denoted as a backbone, and a proposal head is used to create a large number of prior region proposals. In the second stage, the prior proposals are processed by a detection head that can assign a label and a series of more accurate coordinates to a box. The structure of the multistage network is shown in Fig. 3(b). This is an extension of the two-stage detector. A single detection head can only perform limited recognition and regression. This kind of detection head focuses on increasing the quality of proposals for high-quality object detection, as high-quality prior proposals are not always enough to train an excellent detector. In this network, a cascade regressor $f(I, b)$ is implemented, which is composed of a series of specialized detection heads

$$f(I, b_p) = f_T \circ f_{T-1} \circ \cdots \circ f_1(I, b) \tag{9}$$

where $T$ is the total number of the cascade stages. The main idea is that each regressor $f_t$ is optimized for the bounding box distribution, and $b_t$ is generated by the previous regressor, rather than the first input box distribution $b_1$. In this way, the proposals are progressively improved, and then, the input distributions for each detector head can contain many more high-quality positive samples.

At each stage $t$, the detector head consists of a classifier $c_t$ and a regressor $r_t$ that are applied to optimized the corresponding IoU threshold $s^t$, where $s^t > s^{t-1}$. Each of them are learned by loss in (3). The multistage network for object detection in HRRS images is applied to solve the problem of insufficient high-quality samples for the training of an ideal detector. Its advantages are summarized as follows.

1) The potential problem of overfitting for large IoU thresholds $u$ is reduced since positive examples become sufficient in all stages.

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TABLE I
CHARACTERISTICS OF THE THREE USED BENCHMARK DATA SETS

| Dataset Name | Categories | Total Images | Instance | Image width | Data Source | Resolution | Annotation | Characteristics |
|---|---|---|---|---|---|---|---|---|
| NWPU VHR10 [21] | 10 | 800 | 3,775 | ∼1,000 | Google Earth | 0.3 - 2 m | horizontal BB | multiple object categories; clean background |
| HRRSD [6], [32] | 13 | 21,761 | 55,740 | ∼11,000 | Google Earth | 0.15 - 1.2 m | horizontal BB | more categories; many instances |
| DIOR [22] | 20 | 23,463 | 192,472 | 800 | Google Earth | 0.5 - 30 m | horizontal BB | complex background; massive categories and objects; noise, blur, and occlusion |

2) Detectors in the deep stages are optimal with high-quality samples and also enhance the performance of the whole network.

## IV. EXPERIMENTS AND ANALYSIS

In this section, we review the extensive experiments to evaluate the performance of the proposed detector. The experimental data sets, settings, and evaluation metrics are introduced first. Then, a sensitivity analysis experiment is involved to evaluate the influence of the parameters of the proposed detector on its performance and select a set of appropriate parameters. Next, an experiment of ablation study is performed to analyze the performance and computation cost of each component in the detector. Finally, our detectors are compared with the existing milestone detectors on three public data sets: NWPU VHR-10, HRRSD, and detection in optical remote (DIOR).

### A. Experimental Data Sets, Settings, and Evaluation Metrics

*1) Experimental Data Sets:* There are three data sets used in this study: one small-scale data set, i.e., the Northwestern Polytechnical University Very High Resolution 10-Class (NWPU VHR-10) data set [21], and two large-scale data sets, i.e., HRRSD [6] and object DIOR sensing [22]. Their attributes are listed in Table I, and they are described in detail later.

*NWPU VHR-10 [21]:* It is an available ten-class geospatial object detection data set created from Northwestern Polytechnical University, Xi'an, China, used for research purposes. These ten object classes are AL, storage tank (ST), SH, tennis court (TC), baseball diamond, harbor, bridge, basketball court (BC), ground track field, and VE. This data set has a total of 800 very-high-resolution (VHR) images that were cropped from Google Earth [1] and the Vaihingen data set and then manually annotated into 3775 instances.

*HRRSD [6], [32]:* This data set is a large-scale data set with 21 761 RGB images extracted from Google Earth having spatial resolutions from 0.15 to 1.2 m. It can be considered as an extended version of the NWPU VHR-10 data set with increasing the category number of objects to 13, and additional classes are crossroad, T junction, and parking lot. The data set is balanced for each class, and each class has 3700–5000 instances. Due to the massive object instances and sufficient appearance variations, this data set is a real challenge in the remote sensing community.

*DIOR [22]:* The DIOR data set is a more large-scale data set for object detection. It contains 23 463 images at $800 \times 800$ pixels and 192 472 instances labeled with horizontal bounding boxes. The images are collected from Google Earth and have resolutions ranging from 0.5 to 30 m. This data set has sufficient variations of scale, weather, seasons, imaging conditions, and quality, in addition to high levels of interclass similarities and intraclass diversity. It is also one of the larger-scale data sets, with massive images and tremendous numbers of object instances.

*2) Experimental Settings:* To assess the detection algorithms, data sets should be split into a training set and a test set. For the NWPU VHR-10 data set, we randomly divided 60% for training and 40% for testing according to the settings of this work [79]. For the other two data sets, we used the official training set and test set. In this study, a shallow-feature-based method, spatial sparse coding bag-of-words (SSCBOW) [80], and a series of deep-learning-feature-based methods, including YOLO [10], [61], [62], SSD [7], rotation-invariant convolutional neural network (RICNN) [81], Faster R-CNN [59], R-FCN [60], and multi-scale (MS)-CNN [82], were adopted for comparison.

All experiments were carried out on a workbench equipped with an Intel CPU i7 5820k, two NVIDIA RTX Titan GPUs, and 32-GB DDR4 memory. The operating system was Ubuntu 18.04. The batch size was set to 3 for each GPU. All the DL models were constructed based on the Multimedia Laboratory (MMLab) Detection Toolbox and Benchmark (MMDetection) [83] toolbox and the Pytorch [84] framework. The entire training process was 40 epochs for the NWPU VHR-10 and 30 epochs for the HRRSD and DIOR data sets. The initial learning is set to 0.01. The number of detection heads in our multistage detector was set to 3 by default. All the experiments were performed five times by averaging overall results.

*3) Evaluation Metrics:* Detection accuracy is always evaluated by precision and recall. Precision measures the percentage of the prediction results that are correct. Recall is the fraction of the total number of positive instances that were actually predicted. Precision and recall can drive average precision (AP), which is the metric most used in recent research. AP is usually computed for each class separately. Therefore, the mean AP (mAP), which is the average of the AP values of all object categories, has been adopted as the final measure for evaluating overall accuracy [85]–[87].

The detection speed is evaluated by frames per second (FPS) and training time per batch. FPS is a metric to express how

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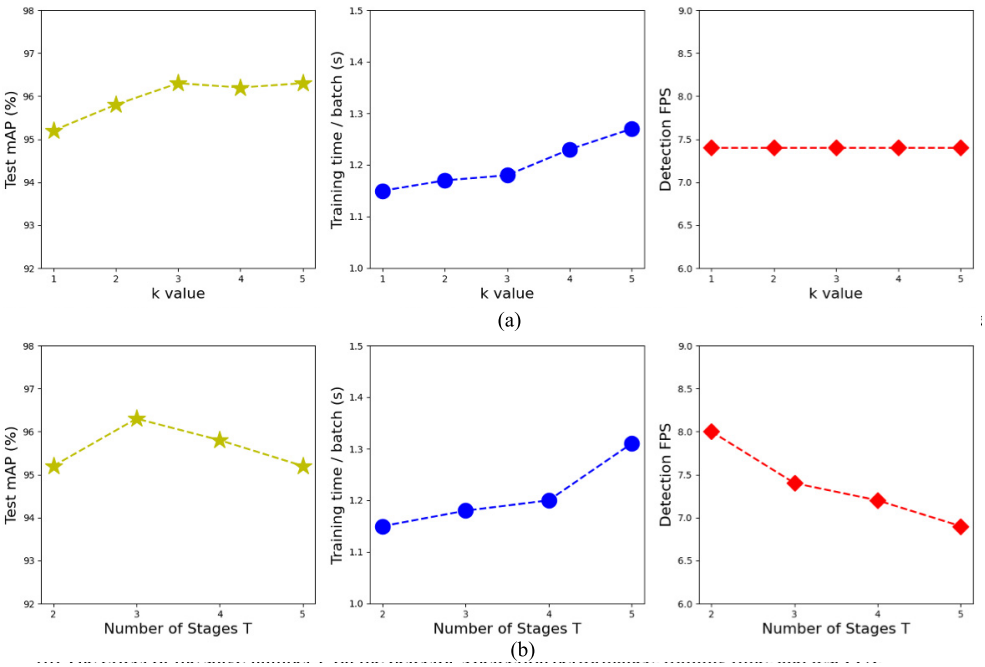



Fig. 4. Experimental results of sensitivity analysis for $k$ in IoU balance sampling strategy and the number of stages $T$. (a) Effect of different $k$ of the IoU balance sampling strategy on the model's detection performance, training time, and test FPS. (b) Effect of stage number $T$ on the detector's detection performance, training time, and test FPS.

fast the detector is. It means the number of image FPS that the detector can process. Training time per batch is the average training time of each iteration for the detectors.

### B. Sensitivity Analysis

In the experiment, there are two parameters, the divided group number of proposal candidates, $k$, in the IoU balance sampling strategy, and the number of detection heads $T$ in the proposed SB-MSN detector. We separately evaluate their effects on the overall performance, training time per batch, and detection FPS.

*1) Sensitivity Analysis of $k$:* The sensitivity analysis of the impact of $k$ on the detector's performance is evaluated first on the NWPU VHR-10 data set. The values of $k$ range from 1 to 5. Fig. 4(a) shows the experimental results. The overall detection accuracy slightly increases with the increment of $k$ and does not change obviously after $k$ greater than 3. As for the computation cost, the average training time per batch continuously rises from 1.21 to 1.3 s. Meanwhile, the detection FPS is insensitive to $k$. Therefore, the increase of $k$ would cause additional calculation for the detector.

*2) Sensitivity Analysis of $T$:* Fig. 4(b) shows the effect of the stages' number $T$ on the detector's accuracy and speed. The values of $T$ ranges from 2 to 5. It can be seen that, when $T$ increases from 2 to 3, the detector can achieve a great improvement in detection accuracy by about 1.2% mAP. However, when it increases from 3 to 5, the overall accuracy continuously decreases. A possible reason is that there are fewer training proposals available and may cause overfitting with more and more detection stages. In addition, the increment of the number of detection head stages would significantly increase the training time per batch and decrease the detection FPS.

Based on the abovementioned experimental results, $k = 3$ and $T = 3$ are relatively reasonable experimental settings, which are used in the subsequent experiments.

### C. Ablation Study

The second experiment is an ablation study, which is used to evaluate the effect of various components in the proposed SB-MSN on the overall performance and computational cost. Faster RCNN, as a standard two-stage model, is used as the baseline model. Because our SB-MSN was constructed based on the multistage Cascade RCNN [77], [78], it was set as a comparison method and different component added on it to evaluate performance changes. ResNeXt 101 [55] was used as the backbone for all detectors in this experiment.

The experimental results are shown in Table II, where it can be seen that the single multistage detector achieved an mAP of 93.2%, which is better than Faster RCNN. In addition, the detection performance had a great improvement of about 1.6% when the multiscale information retention module was added. The IoU balanced sampling and the balanced L1 loss further boosted the detection performance at 0.9% and 0.6% mAP, respectively. The experimental results prove that the various components of this model are effective at improving detection accuracy. For the computational cost, the several components except the balance L1 loss function increase the training time per batch and decrease the detection FPS. The balance L1 loss function can slightly save training time compared with the smooth L1 loss.

### D. Experimental Results and Analysis on NWPU VHR-10 Data Sets

In the third experiment, we compared the detection performance of our model with a number of state-of-the-art methods

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TABLE II
ABLATION STUDY OF SB-MSN ON THE NWPU VHR-10 DATA SET

| Method | Mluti-stage Detection Head | Multi-scale information retention module | IoU-balanced sampling strategy | Balance L1 loss | mAP(%) | training time / batch (s) | prediction FPS |
|---|---|---|---|---|---|---|---|
| Faster RCNN [59] (Base model) | | | | | 92.2 | 0.86 | 10.7 |
| Cascade RCNN [77], [78] | √ | | | | 93.2 | 0.92 | 9.2 |
| SB-MSN | √ | √ | | | 94.8 | 1.17 | 7.5 |
| SB-MSN | √ | √ | √ | | 95.7 | 1.23 | 7.4 |
| SB-MSN | √ | √ | √ | √ | 96.3 | 1.18 | 7.4 |

TABLE III
EXPERIMENTAL RESULTS ON THE NWPU VHR-10 DATA SET*

| Method Name | PL(%) | SH(%) | ST(%) | BD(%) | TC(%) | BC(%) | GTF(%) | HA(%) | BR(%) | VE(%) | mAP(%) |
|---|---|---|---|---|---|---|---|---|---|---|---|
| SSCBOW [80] | 50.6 | 50.8 | 33.4 | 43.5 | 33.3 | 14.9 | 10.1 | 58.3 | 12.5 | 33.6 | 30.2 |
| Faster RCNN [59] (VGG16) | 82.8 | 77.5 | 52.5 | 96.3 | 62.9 | 68.8 | 98.4 | 82.5 | 78.8 | 63.8 | 76.4 |
| Faster RCNN [59] (ResNeXt 101) | 99.7 | 83.7 | 94.5 | 94.0 | 84.3 | 99.9 | 90.2 | 87.7 | 90.6 | 97.5 | 92.2 |
| YOLOv1 [10] | 60.1 | 62.7 | 28.8 | 85.7 | 58.4 | 82.2 | 88.7 | 75.1 | 72.5 | 52.3 | 66.7 |
| YOLOv2 [61] | 87.3 | 84.7 | 42.7 | 93.1 | 65.7 | 85.5 | 97.1 | 80.5 | 90.0 | 70.8 | 79.7 |
| YOLOv3 [62] | 99.9 | 85.4 | 91.4 | 96.1 | 93.7 | **100.0** | 93.3 | 72.3 | 89.6 | 97.1 | 91.9 |
| SSD512 [7] | 90.4 | 60.9 | 79.8 | 89.9 | 82.6 | 80.6 | 98.3 | 73.4 | 76.7 | 52.1 | 78.4 |
| RICNN [81] | 88.4 | 77.3 | 85.3 | 88.1 | 40.8 | 58.5 | 86.7 | 68.6 | 61.5 | 71.1 | 72.6 |
| R-FCN [60] | 96.1 | **98.3** | 72.5 | **99.4** | 90.7 | 97.8 | **99.0** | 92.5 | 93.4 | 88.4 | 92.8 |
| MS-CNN [82] | 98.7 | 84.3 | 69.3 | 98.4 | 81.5 | 96.0 | 98.4 | 89.5 | **96.8** | 73.0 | 88.6 |
| Cascade RCNN [77], [78] | 99.7 | 87.3 | 84.0 | 95.6 | 96.7 | 99.9 | 91.1 | 90.6 | 89.5 | **98.3** | 93.2 |
| MSOPN-AODN [79] | 99.7 | 89.0 | 80.1 | 98.1 | 86.1 | 92.5 | 98.6 | 94.6 | 94.4 | 82.2 | 91.5 |
| MSOPN-AODN Fuse [79] | 98.7 | 97.2 | 80.6 | 99.1 | 97.3 | 97.1 | 98.7 | **97.2** | 92.7 | 89.2 | 94.9 |
| Our Method | **100.0** | 96.1 | **95.2** | 98.5 | **98.9** | 99.8 | 95.5 | 87.9 | 92.5 | 98.2 | **96.3** |

* Airplane (PL), Ship (SH), Storage tank (ST), Baseball diamond (BD), Tennis court (TC), Basketball court (BC), Ground track field (GTF), Harbor (HA), Bridge (BR), and Vehicle (VE).

to test the superiority of our method. The comparison methods included computer vision methods, such as Faster RCNN [59], YOLO [10], [62], SSD [7], and MS-CNN [82], and remote sensing methods, such as SSCBOW [80], RICNN [81], and MS object proposal network-accurate object detection network (MSOPN-AODN) [79]. All the detection results are shown in Table III and Fig. 5.

In Table III, it can be observed that the shallow-feature-based SSCBOW method had the worst performance, with an mAP of 30%. The mAP values of the remaining deep-learning-based methods all exceeded 70%, which shows that DL methods are more efficient in extracting valuable features of the objects of interest from HRRS images. Among the mainstream computer vision frameworks, the two-stage models with a region proposal generation step, such as Faster RCNN [59] and R-FCN [60], are generally better than the single-stage detectors that directly locate and recognize objects in a single CNN, such as SSD [7] and YOLO [10]. R-FCN and Cascade RCNN achieved the best detection accuracy in mainstream computer vision frameworks with mAP accuracies of 92.8% and 93.2%, respectively. MSOPN-AODN achieved the best accuracy, with 94.8% in all comparison methods, proving that methods that adapt to the characteristics of remote sensing images are more effective than the methods directly transferring the computer vision detectors. The proposed SB-MSN detector achieved 96.3% and significantly increased the accuracy by 1.4%. It achieved 100% in the AL category and remarkably improved the detection accuracy in the typical classes, including TC, BC, and VE.

Partial detection results are shown in Fig. 5. where it can be seen in (a), (d), and (e) that densely distributed targets were detected, while a few targets were missed. In Fig. 5(b), the category of SHs is small-scale compared with the sea and is, therefore, easily missed. Our method is available to address this problem. In Fig. 5(c) and (f), multiclass objects with different scale, color, and texture information, such as ST and bridge, are well taken into account. The proposed method reached the best mAP value and outperformed the second-best result by about 1.4%. The experimental results prove that the proposed SB-MSN is beneficial for object detection in the HRRS image.

### *E. Experimental Results and Analysis on HRRSD Data Set*

In the next experiments, the proposed detector is evaluated using a large-scale data set, HRRSD [6], [32], with a series of milestones for verifying detection performance. All detectors were trained and tested according to the training set provided by the creators [6]. Table IV shows all the detection accuracy values, and Fig. 6 shows some detection examples.

Large differences of scale and aspect ratio, sufficient variation, and many classes in the recently released HRRSD

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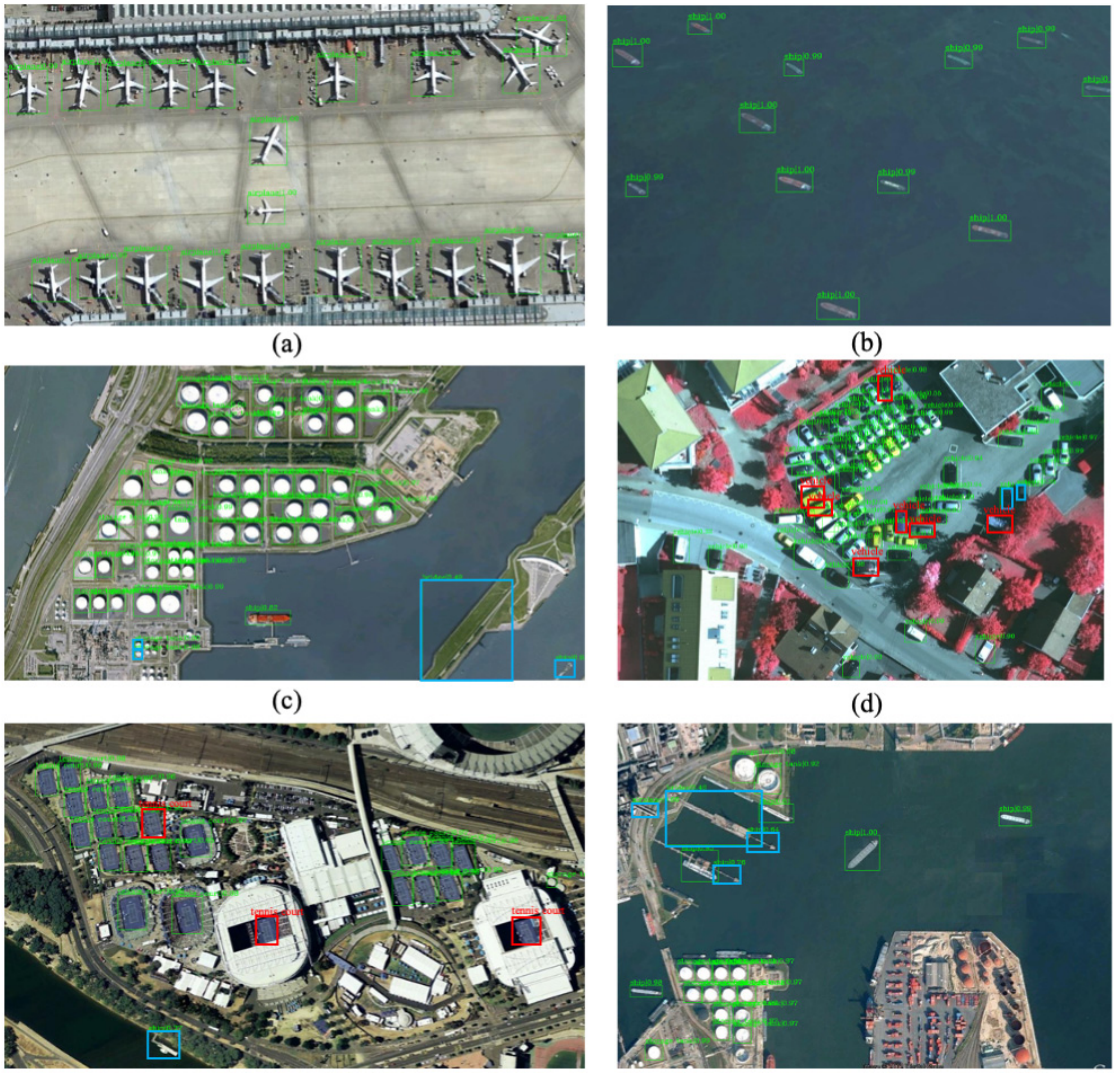



Fig. 5. Some prediction examples of the proposed method on the NWPU VHR-10 data set (Green boxes are the correct predictions. Blue boxes are the false predictions. Red boxes are the missing predictions).

TABLE IV
EXPERIMENTAL RESULTS ON THE HRRSD DATA SET**

| Method Name | PL(%) | SH(%) | ST(%) | BD(%) | TC(%) | BC(%) | GTF(%) | HA(%) | BR(%) | VE(%) | CR(%) | TJ(%) | PK(%) | mAP(%) |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| SSCBOW [80] | 52.3 | 37.6 | 26.3 | 35.0 | 12.2 | 7.9 | 7.5 | 55.5 | 9.3 | 30.5 | 11.6 | 0.8 | 6.8 | 22.9 |
| RICNN [81] | 78.1 | 56.5 | 80.9 | 59.6 | 66.4 | 23.0 | 78.0 | 47.8 | 27.4 | 52.0 | 26.6 | 9.3 | 20.5 | 48.2 |
| YOLOv2 [61] | 84.6 | 78.5 | 72.4 | 62.2 | 67.6 | 41.3 | 94.4 | 74.4 | 79.0 | 65.1 | 43.4 | 46.8 | 45.8 | 65.8 |
| YOLOv3 [62] | 98.1 | 89.9 | 95.7 | 89.0 | 93.8 | 73.0 | 98.1 | 93.8 | 88.0 | 94.2 | 91.7 | 78.3 | 62.3 | 88.2 |
| SSD512 [7] | 98.7 | 79.9 | 94.9 | 83.0 | 88.4 | 56.2 | 97.4 | 85.0 | 83.3 | 79.1 | 90.5 | 70.5 | 43.5 | 80.8 |
| Fast RCNN [9] | 83.3 | 75 | 79.8 | 83.6 | 75.0 | 36.7 | 90.0 | 76.0 | 75.1 | 46.1 | 67.1 | 39.2 | 37.5 | 66.5 |
| GACL Fast RCNN [32] | 85.1 | 74.3 | 80.4 | 82.6 | 77.0 | 42.1 | 89.6 | 78.4 | 76.7 | 50.7 | 68.7 | 38.8 | 39.5 | 68.0 |
| Faster RCNN [59] | 90.8 | 88.5 | 88.7 | 86.9 | 80.7 | 47.9 | 90.6 | 89.4 | 85.5 | 84.0 | 88.6 | 75.1 | 63.3 | 81.5 |
| GACL Faster RCNN [32] | 90.8 | 88.5 | 89.2 | 87.2 | 80.8 | 49.7 | 90.7 | 89.7 | 85.6 | 86.9 | 88.2 | 75.0 | 65.3 | 82.1 |
| Cascade RCNN | 98.0 | 91.4 | 95.6 | 91.3 | 92.6 | 72.2 | 98.2 | 93.9 | 89.6 | 95.5 | 92.3 | 79.7 | 68.4 | 89.1 |
| Ours | **99.3** | **94.5** | **97.8** | **93.9** | **96.0** | **80.2** | **99.0** | **96.0** | **92.7** | **96.6** | **95.3** | **86.4** | **76.5** | **92.6** |

** Airplane (PL), Ship (SH), Storage tank (ST), Baseball diamond (BD), Tennis court (TC), Basketball court (BC), Ground track field (GTF), Harbor (HA), Bridge (BR), Vehicle (VE), Crossroad (CR), T junction (TJ) and Parking lot (PK).

data set make it challenging for DL challenges to achieve satisfactory performance. The DL-based comparison methods achieved detection accuracies of less than 90%. Gated and axis-concentrated localization network (GACL) Fast RCNN and GACL [6], [32], which were designed by the data set creators to address its problems, achieved mAP values of 68% and

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Fig. 6. Some prediction examples of the proposed method on the HRRSD data set (Green boxes are the correct predictions. Blue boxes are the false predictions. Red boxes are the missing predictions).

TABLE V
EXPERIMENTAL RESULTS ON THE DIOR DATA SET***

| Method Name | PL | AI | BF | BC | BR | CH | DA | ESA | ETS | GS | GTF | HA | OP | SH | SD | ST | TC | TS | VE | WM | mAP(%) |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| RCNN [80] | 35.6 | 43.0 | 53.8 | 62.3 | 15.6 | 53.7 | 33.7 | 50.2 | 33.5 | 50.1 | 49.3 | 39.5 | 30.9 | 9.1 | 60.8 | 18.0 | 54.0 | 36.1 | 9.1 | 16.4 | 37.7 |
| RICNN [81] | 39.1 | 61.0 | 60.1 | 66.3 | 25.3 | 63.3 | 41.1 | 51.7 | 36.6 | 55.9 | 58.9 | 43.5 | 39.0 | 9.1 | 61.1 | 19.1 | 63.5 | 46.1 | 11.4 | 31.5 | 44.2 |
| Faster R-CNN (VGG) [59] | 53.6 | 49.3 | 78.8 | 66.2 | 28.0 | 70.9 | 62.3 | 69.0 | 55.2 | 68.0 | 56.9 | 50.2 | 50.1 | 27.7 | 73.0 | 39.8 | 75.2 | 38.6 | 23.6 | 45.4 | 54.1 |
| Faster RCNN with FPN [59] | 54.1 | 71.4 | 63.3 | 81.0 | 42.6 | 72.5 | 57.5 | 68.7 | 62.1 | 73.1 | 76.5 | 42.8 | 56.0 | 71.8 | 57.0 | 53.5 | 81.2 | 53.0 | 43.1 | 80.9 | 63.1 |
| SSD [7] | 59.5 | 72.7 | 72.4 | 75.7 | 29.7 | 65.8 | 56.6 | 63.5 | 53.1 | 65.3 | 68.6 | 49.4 | 48.1 | 59.2 | 61.0 | 46.6 | 76.3 | 55.1 | 27.4 | 65.7 | 58.6 |
| YOLOv3 [62] | 72.2 | 29.2 | 74.0 | 78.6 | 31.2 | 69.7 | 26.9 | 48.6 | 54.4 | 31.1 | 61.1 | 44.9 | 49.7 | **87.4** | 70.6 | **68.7** | **87.3** | 29.4 | **48.3** | 78.7 | 57.1 |
| RetinaNet [63] | 53.7 | 77.3 | 69.0 | 81.3 | 44.1 | 72.3 | 62.5 | 76.2 | 66.0 | 77.7 | 74.2 | 50.7 | 59.6 | 71.2 | 69.3 | 44.8 | 81.3 | 54.2 | 45.1 | 83.4 | 65.7 |
| CornerNet [88], [88] | 58.8 | **84.2** | 72.0 | 80.8 | **46.4** | 75.3 | 64.3 | **81.6** | **76.3** | **79.5** | 79.5 | 26.1 | 60.6 | 37.6 | 70.7 | 45.2 | 84.0 | 57.1 | 43.0 | 75.9 | 65.9 |
| PANet [89] | 60.2 | 72.0 | 70.6 | 80.5 | 43.6 | 72.3 | 61.4 | 72.1 | 66.7 | 72.0 | 73.4 | 45.3 | 56.9 | 71.7 | 70.4 | 62.0 | 80.9 | 57.0 | 47.2 | **84.5** | 66.1 |
| Cascade RCNN [77], [78] | 53.1 | 71.5 | 61.7 | 83.7 | 39.9 | 74.8 | 50.9 | 54.6 | 48.0 | 61.7 | 70.3 | **53.1** | 55.1 | 71.7 | 55.6 | 60.4 | 78.1 | 54.9 | 41.5 | 74.2 | 61.5 |
| ours | **79.6** | 82.2 | **76.4** | **89.8** | 45.6 | **78.2** | **64.8** | 58.9 | 59.3 | 79.2 | **82.4** | 51.8 | **60.8** | 74.4 | **79.7** | 66.4 | 85.6 | **65.4** | 45.1 | 79.9 | **70.3** |

*** Airplane (PL), Airport (AI), Baseball field (BF), Basketball court (BC), Bridge (BR), Chimney (CH), Dam (DA), Expressway service area (ESA), Expressway toll station (ETS), Golf course (GS), Ground track field (GTF), Harbor (HA), Overpass (OP), Ship (SH), Stadium (SD), Storage tank (ST), Tennis court (TC), Train station (TS), Vehicle (VE), Wind mill (WM).

82.1%, respectively. Cascade RCNN reached the best accuracy among the comparison methods, with an mAP of 89.1%. The proposed method was the only one with an mAP of more than 90%, specifically 92.6%, and it achieved the highest detection accuracy for all 13 categories. Fig. 4 shows that our method can detect targets with different scales, aspect ratios,

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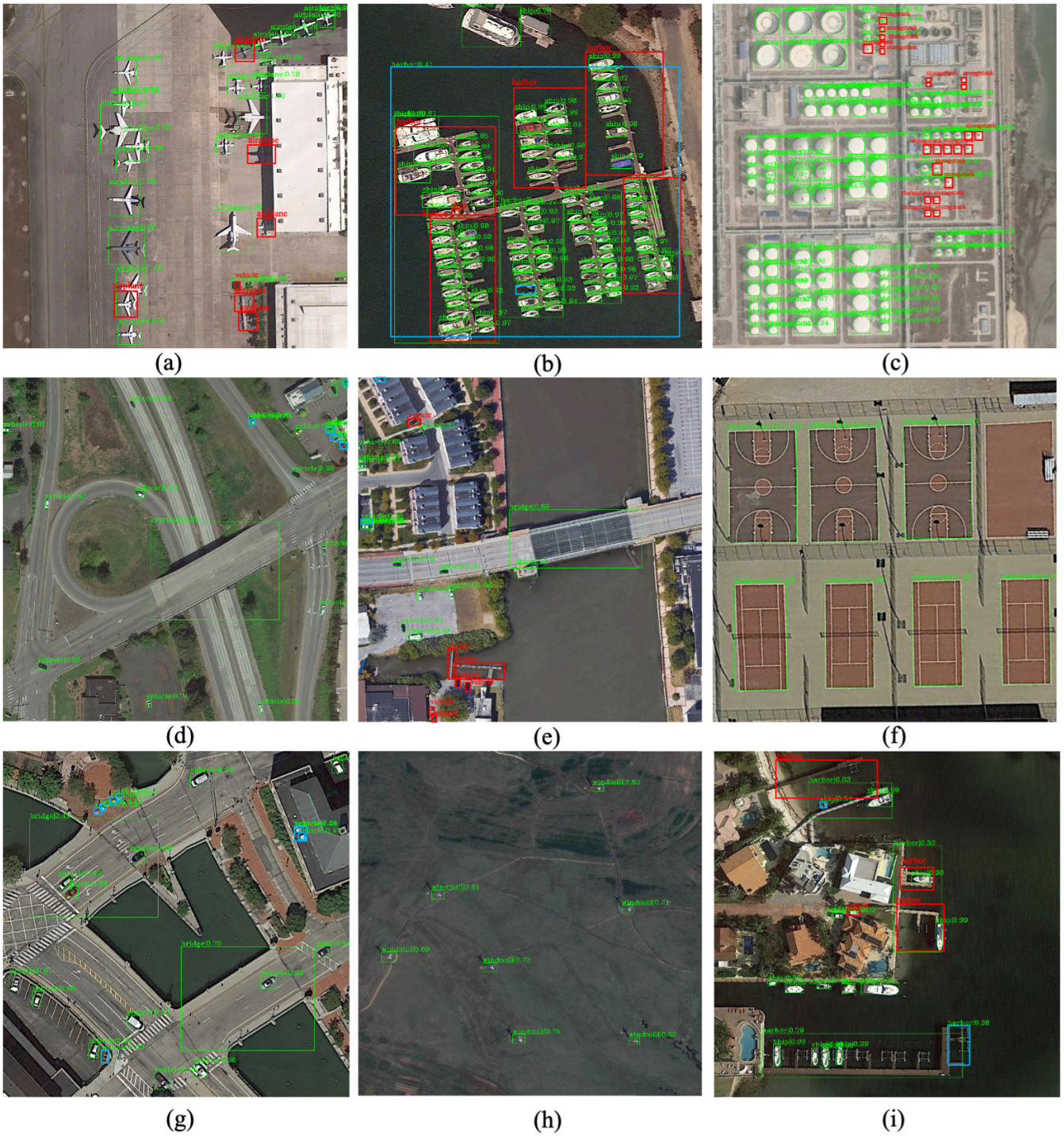



Fig. 7. Some prediction examples of the proposed method on the DOIR data set (Green boxes are the correct predictions. Blue boxes are the false predictions. Red boxes are the missing predictions).

and dense distributions. Compared with the prediction results on NWPU VHR-10, the problems of missed predictions and wrong detections are more severe in this data set. Especially, the proposed model may mistakenly consider abandoned ST as normal ST objects.

### *F. Experimental Results and Analysis on DIOR Data Set*

The last experiment was carried out with the DIOR [22] data set. As with the HRRSD data set, all detectors were trained and tested on the official training set and test set [22]. Table V shows all the detection accuracy values, and Fig. 7 shows some detection examples.

It can be seen in Table V that the performance of all detectors dropped significantly, generally in the mAP range of 50%–60%, which proves the challenging nature of this data set. CornerNet, RetinaNet, and path augmentation network (PANet) are relatively high-accuracy models, with mAP values of 65.7%, 65.9%, and 66.1%, respectively, and they achieved

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the best results in many categories. Our method was the only one that exceeded the mAP of 70%, and it achieved the best results in nine of the total 20 classes. As can be seen in Fig. 7, the target distribution in the images, such as SH or ST, is more densely distributed and complex. Some targets, such as windmill and VE, are very small-scale and weak-feature. These characteristics put forward higher requirements for detection models. Though the proposed detector achieved detection results for multiclass objects in the data set, proving its effectiveness and superiority, many more works should be made to further improve the recognition performance for the DIOR data set.

## V. Conclusion

In this study, focusing on the problems of insufficient high-quality samples for training a detector, our SB-MSN model was proposed for HRRS image object detection. A series of techniques, including a multiscale information retention module, an IoU-balanced sampling strategy, a balance L1 loss, and a multistage detection network, are used to improve the quality of training examples that allowed for accurate detection and avoided overfitting. Extensive experiments were carried out to evaluate the proposed method on three public data sets, and our SB-MSN had improved detection accuracy of about 1.4% on the NWPU VHR-10, 3.5% on the HRRSD, and 4.2% on the DIOR. In the future, we would apply the proposed SB-MSN to detect small, weak, and rotating objects and continue improving its performance.

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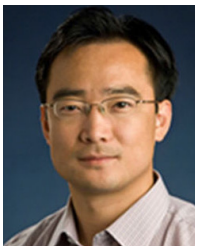
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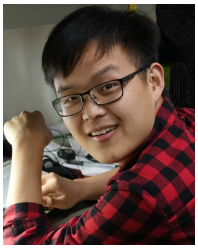
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